

Mac Software Installation

CAREFUL: If you cut-and-paste from a PDF document, likely all non-alphanumeric characters, including commas, apostrophes, quotation marks, underscores, dashes, hyphens, tabs, symbols like ~, and so on will look right but work wrong. This is because typesetting programs use prettier versions of these characters, and the shell doesn't understand anything but simple ASCII characters. Either retype these symbols or just type the whole line by hand.

It is important to log what you do when you're doing system management! Especially log anything that deviates from these instructions.

1 Unix Shell Part 1: Backup

MacOS *is* Unix, so you already have the shell! To access it let's start a Terminal from Spotlight:

Spotlight → terminal.

While you're at it, put the Terminal in your Dock, as you'll be using it a lot. After the Terminal opens, right-click (or equivalent) on Terminal on the Dock, click Options, and click "Keep in Dock". The shell is an interactive programming language. You can control and enhance its interactive behavior! Whenever you start an interactive shell, it first runs several *shell scripts* containing configuration instructions, such as setting up your prompt. These files include `.profile`, `.bash profile`, and `.bashrc` (the Bourne Again SHell Resource file). Which of these files run depends on multiple factors. They are all in your home directory. You may or may not have all of them, and that's ok. First, we'll make a backup of your current shell configurations. Do this **ONLY ONCE**. If you repeat these instructions, do not repeat this step, or you will destroy your backup.

```
cd
mkdir .bash_prePHZ3150
cp -a .bashrc .bash_profile .profile .bash_aliases
.bash_prePHZ3150
```

Don't worry about any "file not found" messages.

To make the minimal changes to the files you already have, we'll put a little bit of code in those, which reads our stuff from a separate file. However, we need to install Python first, as we want our changes to be done after Conda's changes.

2 Miniconda

To install Miniconda, go to: <https://docs.anaconda.com/miniconda/> and download the latest Mac Python 64-bit BASH installer at the top of the page.

In the shell, back up your shell settings file, move to your Downloads directory, run the installer, integrate Miniconda into your shell environment, and update the packages in Miniconda:

```
cd ~/Downloads
bash Miniconda3-latest-MacOSX-x86_64.sh
source ~/.bash_profile
conda update --all
```

Follow the prompts and use the default settings. You will be asked to read and agree to a license. You can scroll with down arrow or page down with the spacebar. At the end, type 'yes' to agree. Press Enter at the next prompt to install as a user (i.e., not administrator). Finally, type 'yes' and press Enter at the last prompt to integrate Miniconda into your environment. The final two commands bring Conda and Python into the current shell environment and update its component packages, respectively.

2.1 Add Packages to Conda

We will use Conda to install some additional packages. You can install individual packages with:

```
conda install <packagename>
```

(This is called metasyntax; you replace <packagename> with the name of the package.)

The packages you want are numpy, scipy, ipython (interactive Python), and matplotlib so type:

```
conda install numpy scipy ipython matplotlib
```

2.2 Spyder

Python differs from a language like C, in that code is not compiled before being run. Rather, it is interpreted by the Python program as it runs. This makes it easy to start coding in a good text editor (like Emacs) and test lines of code in a Python interpreter as you write them. However, some prefer to use an Integrated Development Environment (IDE), which is a program that lets you edit the code, run it, and helps you debug any problems.

Spyder is a well known Python IDE. Install it with:

```
conda install spyder
```

You can run Spyder by typing this to a terminal's shell prompt:

```
spyder &
```

The & puts it "in the background", meaning you will get your shell prompt back right away, before the program exits. We will discuss this in class.

2.3 Jupyter

Jupyter is a python notebook that is very popular in industry. Like Spyder it lets you edit, run and debug a code, but in your favorite web browser. To install it use:

```
conda install jupyter
```

To run Jupyter type:

```
jupyter notebook &
```

to a terminal's shell prompt.

2.4 Customizing the Shell

Now it's time to add the class customizations. Go to:

WebCourses → Files → handouts

Download the files `bash_aliases` and `mac-add-to-bash-inits` to your Downloads folder. Move them to your home directory:

```
mv -i ~/Downloads/bash_aliases ~/.bash_aliases
```

```
mv -i ~/Downloads/mac-add-to-bash-inits ~/
```

Adding a “.” to the front of bash aliases is important! If it asks if you want to overwrite an existing version, say no. This means you already have a `.bash_aliases` file. If it was there before you started the course, you'll need to merge the contents of the two files. Ask course staff if you don't know how. Next, see what shell configuration files you already have:

```
ls .profile .bash_profile .bashrc
```

append the contents of `mac-add-to-bash-inits` to any of the three shell configuration files that you already have (in the listing above, it will say “file not found” for the rest). For example, to append it onto the end of `~/.bash_profile`:

```
cat mac-add-to-bash-inits >> ~/.bash_profile
```

Do this **ONLY** for files that already exist!

The next time you start a shell, you'll have some useful configurations and new commands. We'll talk about these in class. If you want them right away, do:

```
source ~/.bash_aliases
```

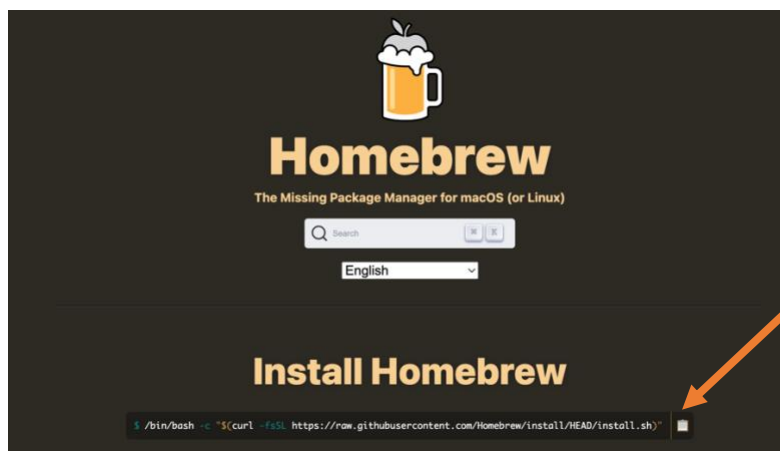
in every terminal.

3

3 Homebrew

To install Homebrew, go to: <http://brew.sh/>

At the top of the page, you will see a line of shell script to paste into a terminal:



Copy and paste that text into a terminal. You may need to supply a password. You now have hundreds of open-source packages available! If you like digital image editing, try GIMP, for example.

4 Emacs

Emacs is primarily a text editor, but to the experienced user is a full IDE (especially considering the various libraries that can be added), as well as just about anything (terminal, window manager, web browser, email reader, etc.). To install, open a terminal and type:

```
brew cask install emacs
```

and follow the prompts. The cask command enables integration with the Mac's window system.

To run, go to your Terminal and type:

```
emacs &
```

From now on, use Emacs (with capital 'E') rather than TextEdit to edit your log:

```
emacs <filename> &
```

5 Git

To install the Git revision control system, open a terminal and run:

```
brew install git
```

and follow the prompts.